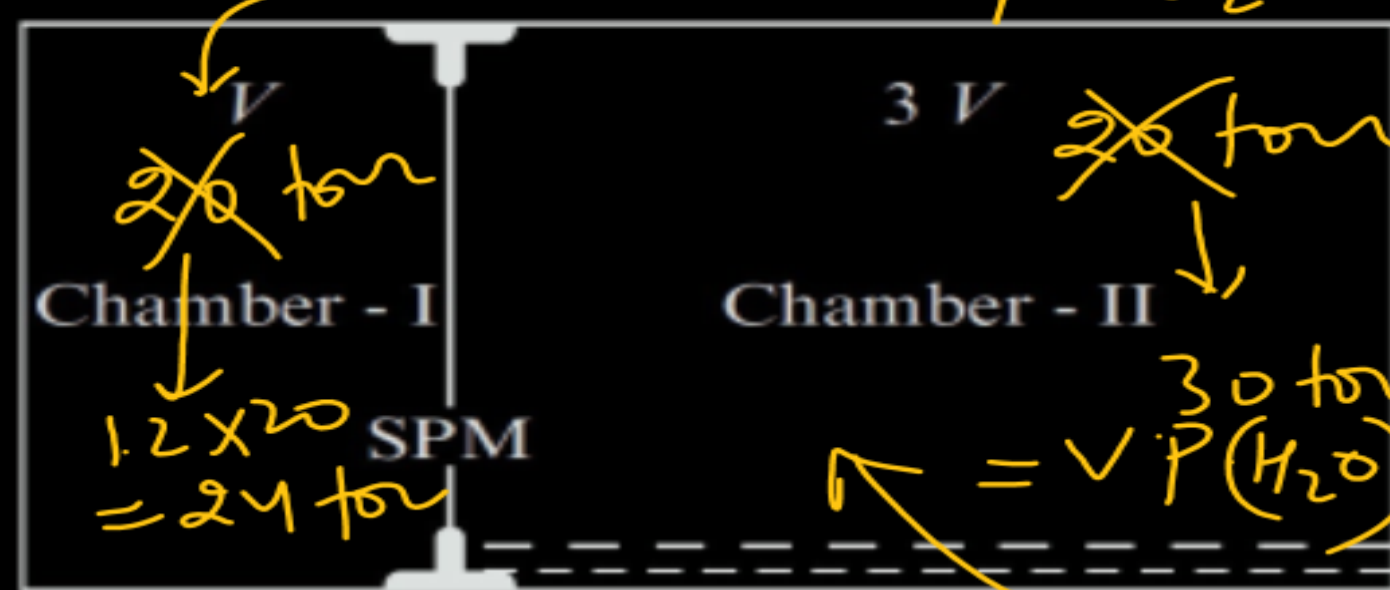


A container is divided into two parts in the ratio 1 : 3, by a fixed semipermeable membrane (SPM), which permits the movement of NO₂ gas only. At T K, Chamber – I contains only N₂O₄ gas at 20 mm Hg pressure and Chamber – II contains small amount of liquid water, but it is sufficient to maintain equilibrium with vapours at any temperature. Now, both the chambers are heated to ' $1.2T$ ' K at which 50% of N₂O₄ is dissociated into NO₂. The vapour pressure of water at T K and ' $1.2T$ ' K are 20 and 30 mm Hg, respectively.



$$p_{N_2O_4} = 12 \text{ } \left. \begin{array}{l} p_{NO_2} = 6 \end{array} \right\} = 18 \text{ tor} \text{ Finally}$$



$$24 - 12 = 12 \text{ tor}$$

$$24 \text{ tor}$$

$$(p_{NO_2})_I = (p_{NO_2})_{II}$$

$$= \frac{1}{4} \times 24$$

$$= 6 \text{ tor}$$

$$(V \cdot P)_{H_2O(l), 1.2T} = 30 \text{ tor}$$

$$(p_{NO_2})_{II} = 6 \text{ tor}$$

$$= 36 \text{ tor}$$

Choose the correct option(s)

(a) final pressure in chamber I is 18mm of Hg

(b) Final pressure in chamber II is 36 tor

(c) Final pressure in chamber I is 15 mm of Hg

(d) Final pressure in chamber II is 25 mm of Hg

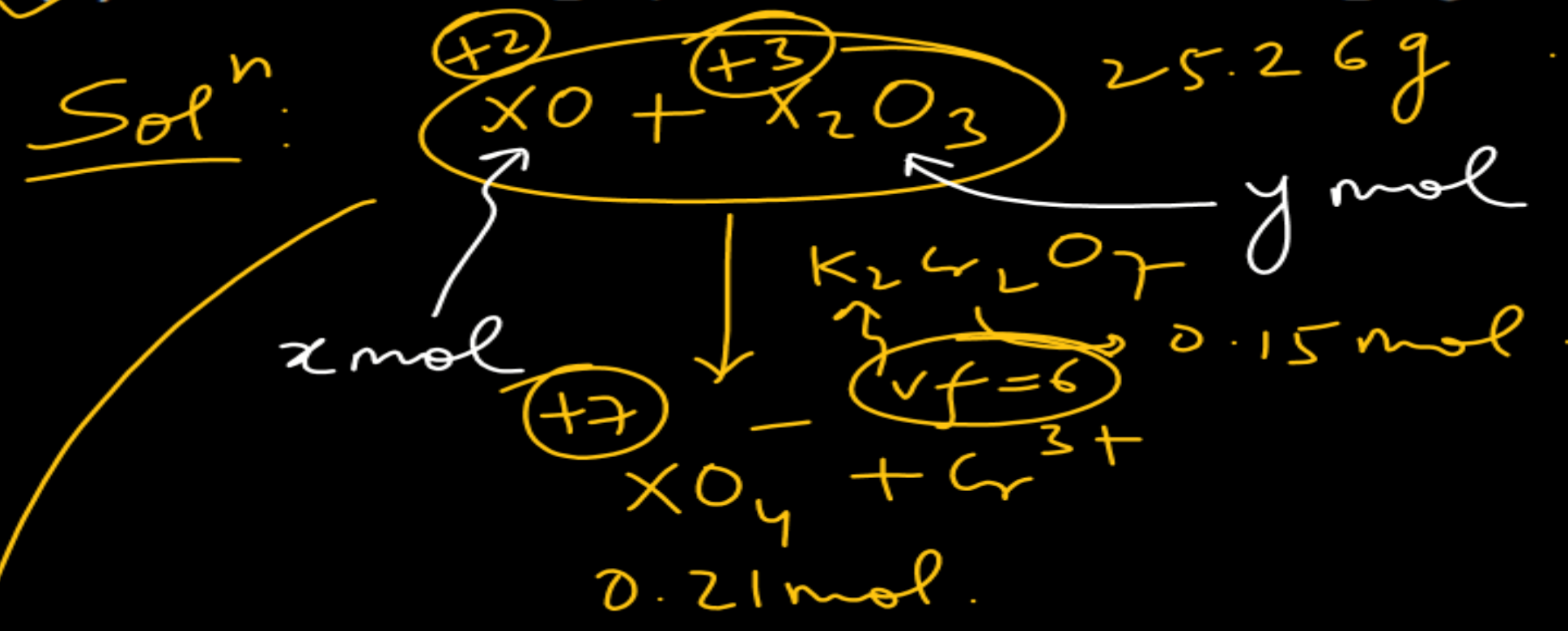
Ans: a, b

Ans: b, c, d ✓

An amount of 0.15 moles of $K_2Cr_2O_7$ is required to oxidize a mixture of XO and X_2O_3 (total mass = 25.26 g)

g) to form XO_4^- and Cr^{3+} . If 0.21 moles of XO_4^- is formed, then the correct information(s) is/are

- (a) Atomic mass of $X = 160$. ✗
 (b) Moles of XO in original mixture = 0.06. ✓
 (c) Moles of $Cr_2O_7^{2-}$ combined with $X_2O_3 = 0.10$. ✓
 (d) Total moles of H^+ consumed = 0.99. ✓



$$n_{XO_4^-} = 0.21 = n_X = x + 2y \quad \text{--- (3)}$$

$$x = 0.9 - 4 \times 0.21 = 0.9 - 0.84 = 0.06 \text{ mol}$$

$$y = \frac{0.21 - 0.06}{2} = 0.075 \text{ mol}$$

$$\begin{array}{r} 48 \\ 7.5 \\ \hline 240 \\ 336 \\ \hline 360.0 \\ \hline 96 \end{array}$$

$$0.06(M_X + 16) + 0.075(2M_X + 48) = 25.26 \quad \text{--- (1)}$$

$$6M_X + 96 + 15M_X + 7.5 \times 48 = 2526$$

$$21M_X = 2526 - 456$$

$$\begin{array}{r} 2526 \\ 456 \\ \hline 2070 \end{array}$$

$$M_X = \frac{2070}{21} \approx 100$$

$$x(M_X + 16) + y(2M_X + 48) = 25.26 \quad \text{--- (1)}$$

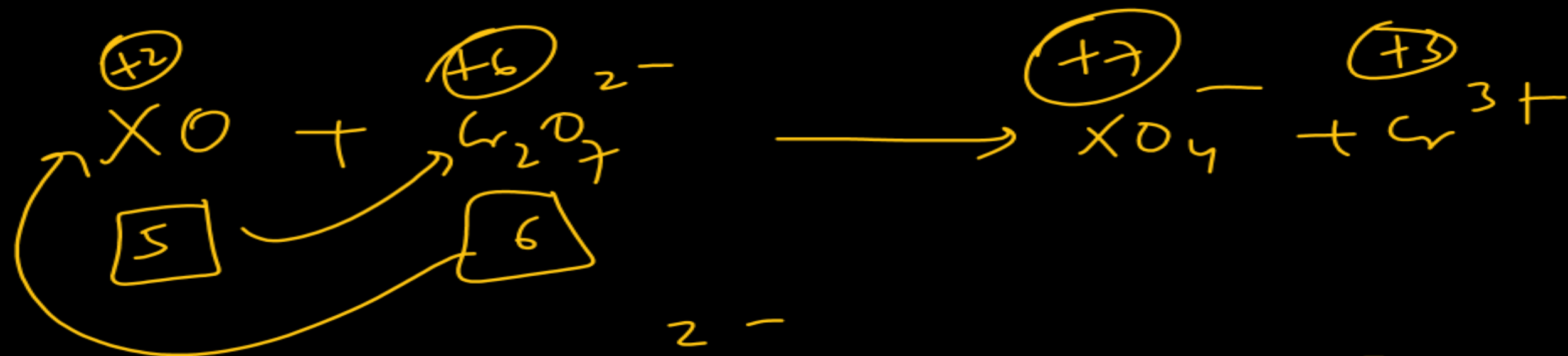
$$x \times 5 + y \times 8 = 0.15 \times 6$$

$$5x + 8y = 0.9 \quad \text{--- (2)}$$

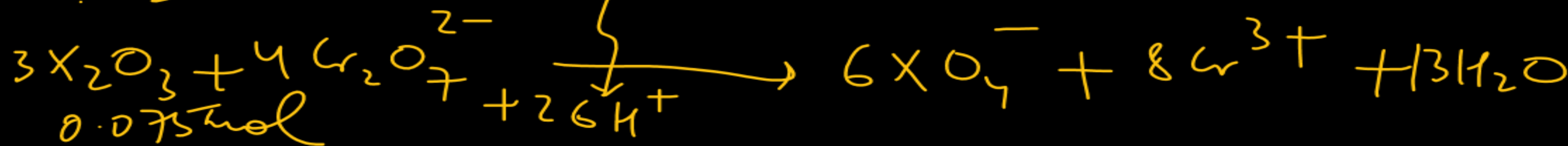
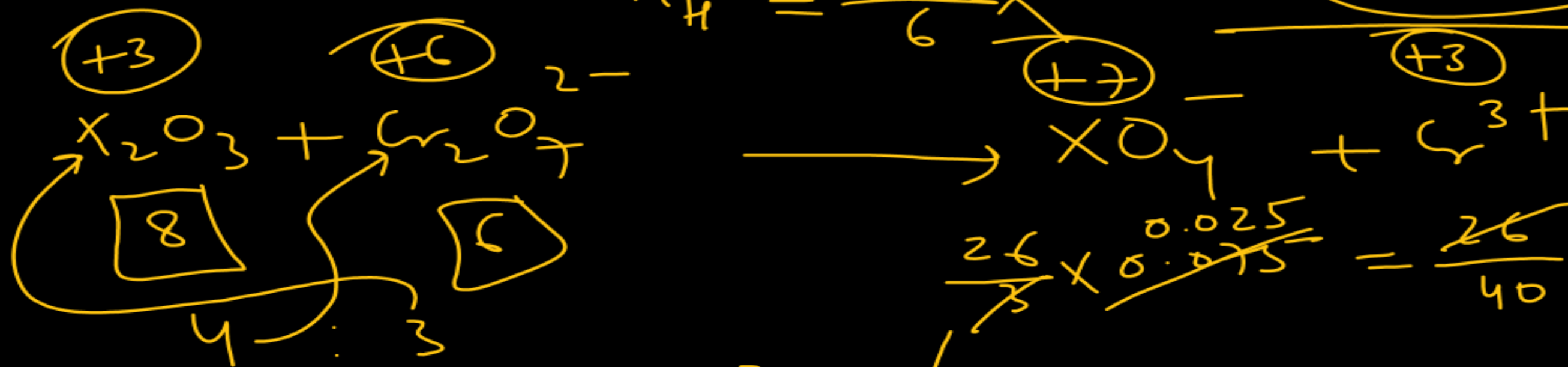
$$x \times 8 = a \times 6$$

$$a = \frac{0.60}{6} = 0.1 \text{ mol}$$

$\rightarrow K_2Cr_2O_7$



$$n_{\text{H}^+} = \frac{34}{6} \times 0.06 = 0.34\text{mol}$$



$$\frac{26}{3} \times 0.075 = \frac{26}{40} = 0.65\text{mol}$$

$$\begin{array}{l}
 (n_T)_{\text{H}^+} \text{ needed} \\
 = 0.34 \\
 \quad 0.65 \\
 \hline
 0.99\text{mol}
 \end{array}$$

Two moles of an ideal gas ($C_{v,m} = 1.5R$) is subjected to the following changes in states.

A (500 K, 5 bar) $\xrightarrow[\text{Expansion}]{\text{Reversible isothermal}}$ B $\xrightarrow[\text{Isochoric cooling}]{\text{500 K, 2 bar}}$ C (250 K, 1 bar) $\xrightarrow[\text{Compression}]{\text{Single stage adiabatic}}$ D (3 bar), 450 K

The correct statement(s) is/are

(a) The pressure at B is 2.0 bar.

(c) $\Delta H_{CD} = 1000 R$

(b) The temperature at D is 450 K.

(d) $\Delta U_{BC} = 375 R$

Solⁿ:

$$5 \times V_A = 2 \times R \times 500$$

$$V_A = 200 R$$

Ans: a, b, c

$$\Delta H_{CD} = n C_p (T_2 - T_1)$$

$$= 2 \times 2.5 R (450 - 250)$$

$$= 5 R \times 200$$

$$= 1000 R$$

$$\Delta U_{BC} = n C_v (T_2 - T_1)$$

$$= 2 \times 1.5 R (250 - 500)$$

$$= 3 R \times (-250) = -750 R$$

in adi-process
(compression)

$$q_{irr} = 0,$$

$$\Delta U = W_{irr}$$

$$n C_v (T_2 - T_1) = -P_{ext} (V_2 - V_1)$$

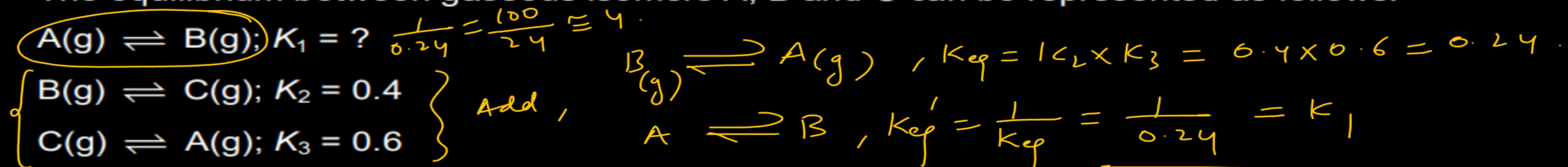
$$2 \times 1.5 R (T_D - 250) = -3 \left(\frac{2 \times R \times T_D}{3} \right)$$

$$3 T_D - 750 = -2 T_D - 2 \times 250$$

$$5 T_D = 250$$

$$T_D = 450 K$$

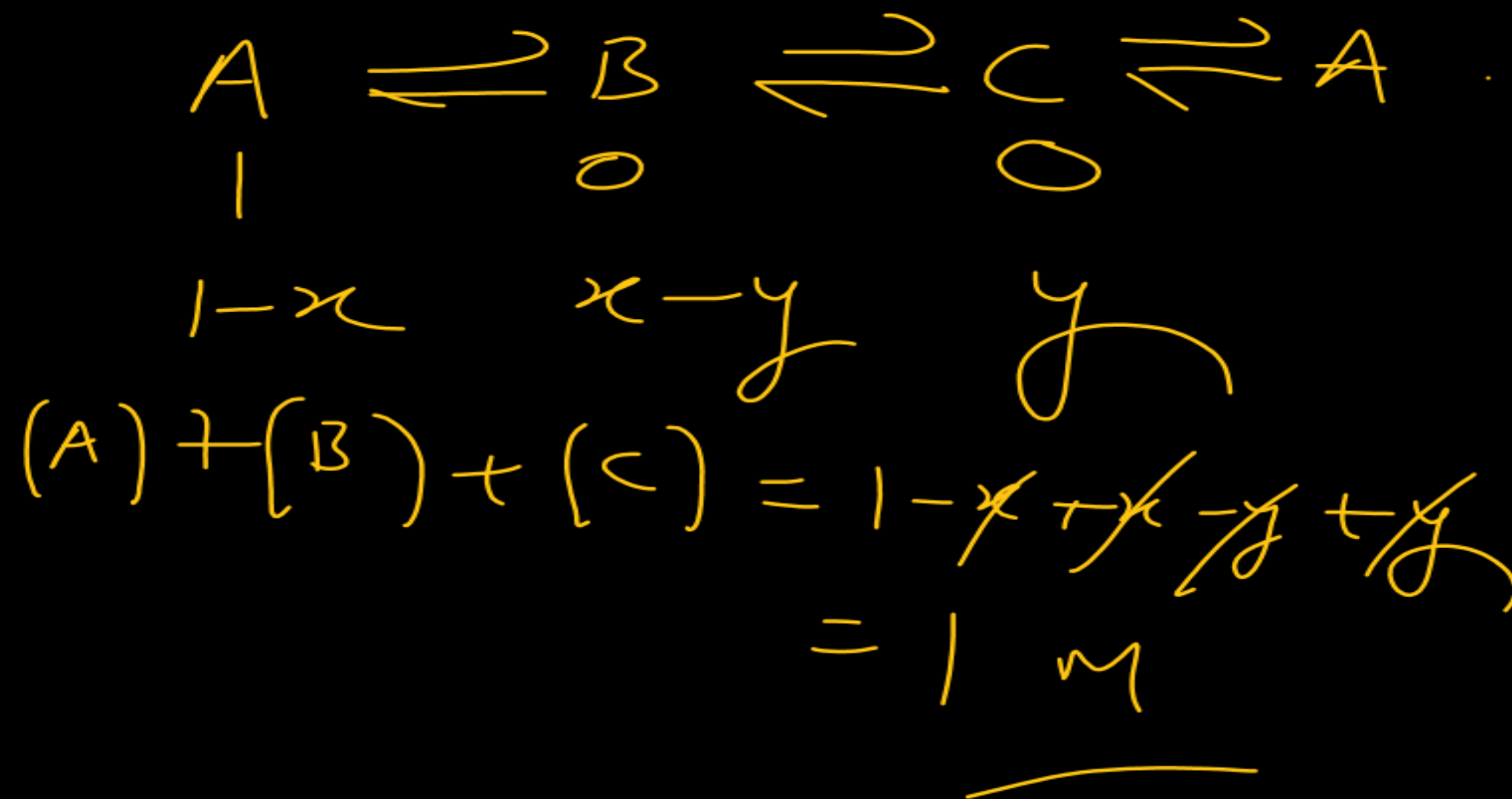
The equilibrium between gaseous isomers A, B and C can be represented as follows.



If one mole of A is taken in a closed vessel of volume 1 L, then

- ☒ (a) $[A] + [B] + [C] = 1$ M at any moment of reaction.
- ☐ (b) concentration of C is 1.2 M at equilibrium.
- ☒ (c) the value of K_1 is $1/0.24$.
- ☒ (d) isomer A is thermodynamically least stable.

$$\underbrace{\Delta G_r^\circ}_{\uparrow} = -RT \ln \underbrace{(K_{eq})}_{-ve} \uparrow$$



Match the columns.

Column I (pH value)

Column II (Solution)

(A) 5.0

(P) pH at which $\text{Mg}(\text{OH})_2$ just start precipitating from $2.0 \times 10^{-3} \text{ M-Mg}^{2+}$ solution (K_{sp} of $\text{Mg}(\text{OH})_2 = 2 \times 10^{-6} \text{ M}^3$).

(B) 9.0

(Q) Maximum pH at which 0.1 mole of $\text{Al}(\text{OH})_3$ dissolves completely as Al^{3+} in 1 L solution (K_{sp} of $\text{Al}(\text{OH})_3 = 1 \times 10^{-28} \text{ M}^4$).

(C) 12.5

(R) pH at which 0.1 M- CH_3COOH dissociates 1000/11% (K_a of $\text{CH}_3\text{COOH} = 1.0 \times 10^{-5} \text{ M}$).

(D) 6.0

(S) pH of 0.001 M-HA (weak acid) solution ($\text{p}K_a$ of HA = $\text{p}K_b$ of A^-).

(T) pH of $3 \times 10^{-5} \text{ M-CaA}_2$ solution (CaA_2 dissociates completely in water; K_a of HA = 5×10^{-9}).

$$\textcircled{S} \text{ pH} = \frac{1}{2} (\text{p}K_a - \log c) = \frac{1}{2} (7 - \log 10^{-3}) = \textcircled{5}$$

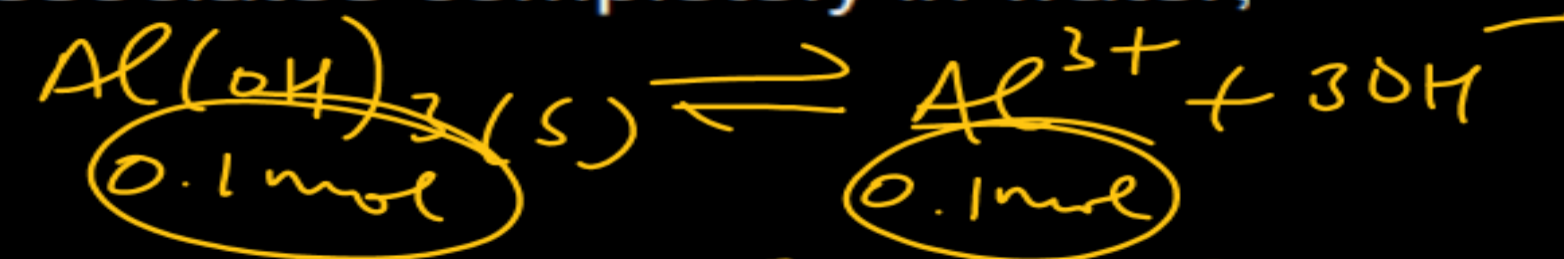
$\textcircled{P}$



$$2 \times 10^{-3} \times (\text{OH}^-)^2 = 2 \times 10^{-6}$$

$$(\text{OH}^-)^2 = 10^{-3} \Rightarrow (\text{OH}^-) = 10^{-1.5} \Rightarrow \text{pOH} = 1.5, \text{pH} = 12.5$$

$\textcircled{Q}$



$$0.1 \times (\text{OH}^-)^3 \geq 10^{-28}$$

$$(\text{OH}^-) = (10^{-27})^{1/3} = 10^{-9} \text{ M}$$

$$\text{pOH} = 9, \text{pH} = 5$$

$$\alpha \text{ appr.} = \sqrt{\frac{10^{-7}}{10^{-3}}} = 0.01 < 0.1$$

$$\text{p}K_a + \text{p}K_b = \text{p}K_w = 14$$

$$\text{p}K_a = 7 \Rightarrow K_a = 10^{-7}$$

(R) $\text{CH}_3\text{COOH} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}^+$
 $0.1 - x$ x $(x + c)$

$$\text{pH} = \frac{1}{2}(14 + 9 - 0.7)$$

$$= \frac{1}{2}(18.0.8)$$

$$= 9.04$$

$$x = \frac{x}{0.1} = \frac{10^{-9.04}}{10^{-1}} = 10^{-8.04}$$

$$x = \frac{1}{11}$$

$$x^2 + 0.2 \times 10^{-5}x - 1.2 \times 10^{-10} = 0$$

$$x = \frac{-0.2 \times 10^{-5} \pm \sqrt{0.04 \times 10^{-10} + 4 \times 1.2 \times 10^{-10}}}{2}$$

$$= \frac{(2.2 - 0.2) \times 10^{-5}}{2}$$

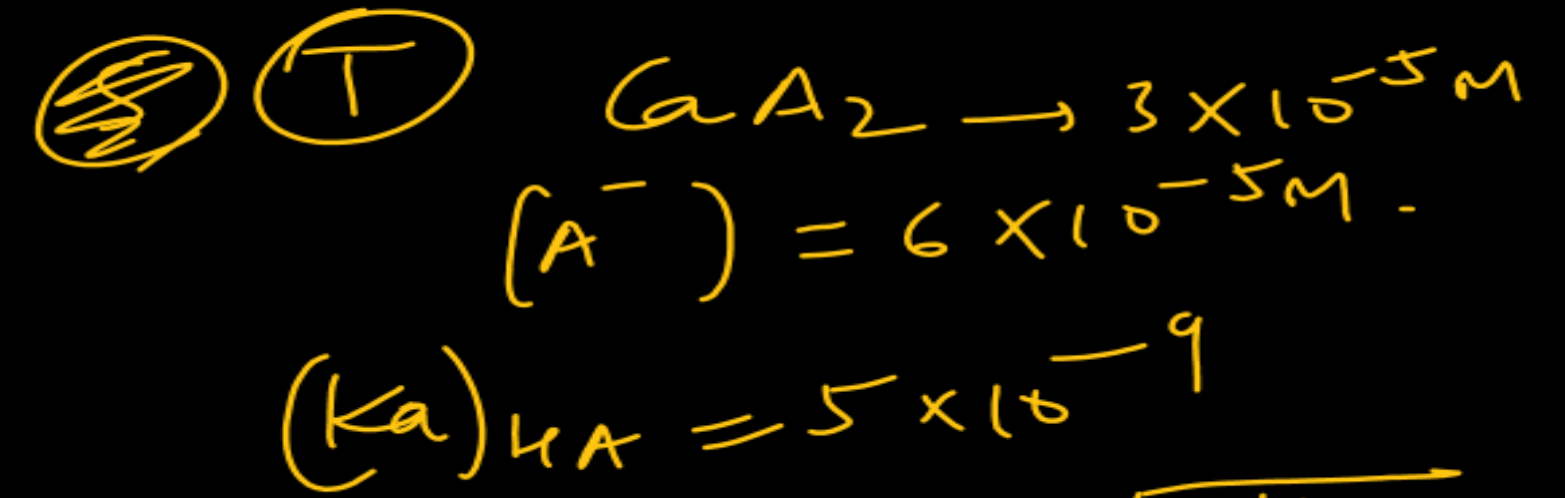
$$\approx 1 \times 10^{-5} \text{ M}$$

$$[\text{H}^+] = 10^{-6} \text{ M}$$

$$\text{pH} = 6$$

$$[\text{OH}^-] \approx 10^{-5} \text{ M}$$

$$\text{pOH} = 5, \text{pH} = 9$$

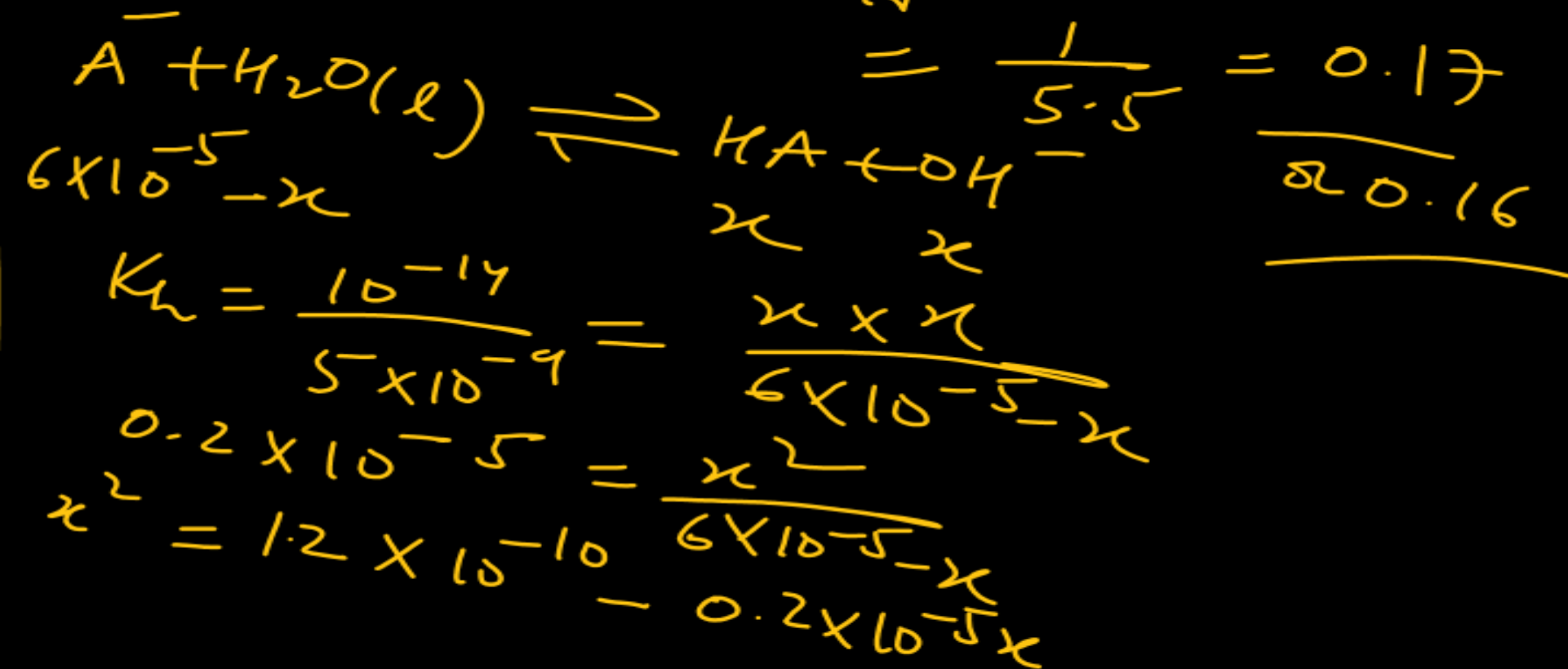


$$h_{\text{appr.}} = \sqrt{\frac{K_h}{c}} = \sqrt{\frac{K_w}{K_a \times c}}$$

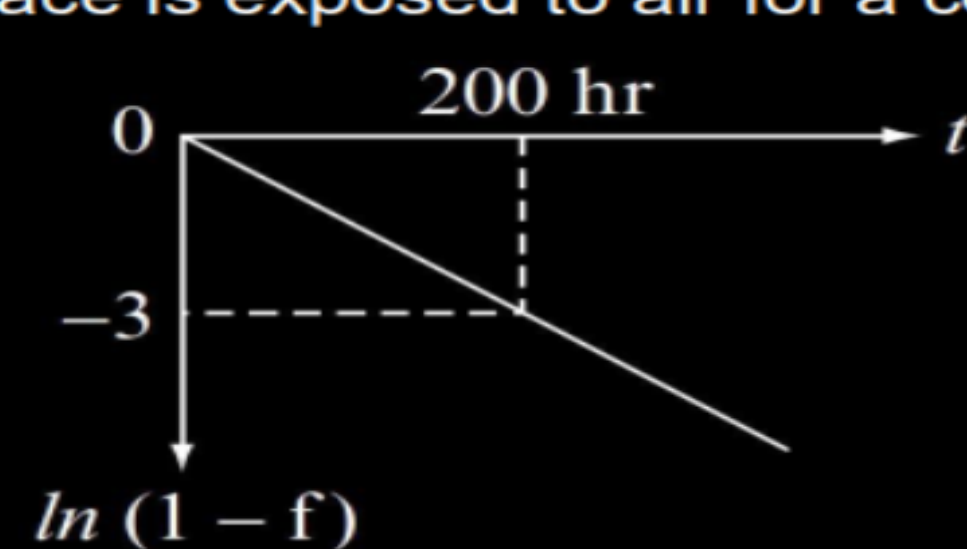
$$= \sqrt{\frac{10^{-14}}{5 \times 10^{-9} \times 6 \times 10^{-5}}}$$

$$= \sqrt{\frac{1}{30}}$$

$$= \frac{1}{5.5} = 0.17$$



The oxidation of metals is generally a slow electrochemical reaction involving many steps. These steps involve electron transfer reactions. A particular type of oxidation involve overall first-order kinetics with respect to fraction of unoxidized metal surface thickness $(1-f)$ relative to the maximum thickness (T) of oxidized surface, when metal surface is exposed to air for a considerable period of time.



Rate law: $\frac{df}{dt} = K(1-f)$,

where, $f = x/T$,

x = thickness of the oxide film at time 't'

and T = thickness of the oxide film at $t = \infty$

For the oxidation of a particular metal, a graph is shown in the figure.

The time taken for thickness to grow 50% of T is

(a) 23.1 h

☒ (b) 46.2 h

(c) 100 h

(d) 92.4 h

$$\int_0^f \frac{df}{(1-f)} = K \int_0^t dt$$

$$-\ln(1-f) = Kt$$

$$K = -\frac{1}{t} \ln(1-f)$$

$$t = 200 \text{ hr}, \ln(1-f) = -3$$

$$K = -\frac{1}{200} \times (-3) = \frac{3}{200} \text{ hr}^{-1}$$

→ for 50% to grow, $x = T/2$

$$f = \frac{1}{2} = \frac{200}{3} \times 0.7 \text{ hr}$$

$$t = -\frac{1}{(3/200)} \times \ln(1 - \frac{1}{2}) = \frac{200}{3} \times 0.7 \text{ hr} = 46.2 \text{ hr}$$

The exponential variation of f with t (h) is

(a) $f = 1 - e^{-3t/200}$

(b) $f = e^{-3t/200} - 1$

(c) $f = e^{-3t/200}$

(d) $f = e^{3t/200}$

$$\frac{df}{dt} = k(1-f)$$

$$-\ln(1-f) = k \times t$$

$$1-f = e^{-kt}$$

$$f = 1 - e^{-\frac{3}{200} \times t}$$

$$f = 1 - e^{-3t/200}$$
